

```
In [1]: !pip install torch torchvision torchaudio scikit-learn medmnist
```

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Requirement already satisfied: torch in /opt/conda/lib/python3.13/site-packages (2.13.0)
Requirement already satisfied: torchvision in /opt/conda/lib/python3.13/site-packages (0.28.0)
Requirement already satisfied: torchaudio in /opt/conda/lib/python3.13/site-packages (2.11.0)
Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.13/site-packages (1.9.0)
Requirement already satisfied: medmnist in /opt/conda/lib/python3.13/site-packages (3.0.2)
Requirement already satisfied: filelock in /opt/conda/lib/python3.13/site-packages (from torch) (3.32.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /opt/conda/lib/python3.13/site-packages (from torch) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in /opt/conda/lib/python3.13/site-packages (from torch) (82.0.1)
Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.13/site-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /opt/conda/lib/python3.13/site-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /opt/conda/lib/python3.13/site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /opt/conda/lib/python3.13/site-packages (from torch) (2026.7.0)
Requirement already satisfied: cuda-toolkit==13.0.3 in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.3.0)
Requirement already satisfied: cuda-bindings<14,>=13.0.3 in /opt/conda/lib/python3.13/site-packages (from torch) (13.3.1)
Requirement already satisfied: nvidia-cudnn-cu13==9.20.0.48 in /opt/conda/lib/python3.13/site-packages (from torch) (9.20.0.48)
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.1 in /opt/conda/lib/python3.13/site-packages (from torch) (0.8.1)
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Requirement already satisfied: nvidia-nccl-cu13==2.29.7 in /opt/conda/lib/python3.13/site-packages (from torch) (2.29.7)

Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in /opt/conda/lib/python3.13/site-packages (from torch) (3.4.5)

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Requirement already satisfied: nvidia-cublas==13.1.1.3.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.1.1.3)

Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.88)

Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.96)

Requirement already satisfied: nvidia-cufft==12.0.0.61.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (12.0.0.61)

Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.3.33)

Requirement already satisfied: nvidia-cufile==1.15.1.6.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (1.15.1.6)

Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.85)

Requirement already satisfied: nvidia-curand==10.4.0.35.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (10.4.0.35)

Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (12.0.4.66)

Requirement already satisfied: nvidia-cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (12.6.3.3)

Requirement already satisfied: nvidia-nvtx==13.0.85.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.85)

Requirement already satisfied: cuda-pathfinder>=1.4.2 in /opt/conda/lib/python3.13/site-packages (from cuda-bindings<14,>=13.0.3->torch) (1.6.0)

Requirement already satisfied: numpy in /opt/conda/lib/python3.13/site-packages (from torchvision) (2.5.1)

Requirement already satisfied: pillow!>=8.3.*,>=5.3.0 in /opt/conda/lib/python3.13/site-packages (from torchvision) (12.3.0)

Requirement already satisfied: scipy>=1.10.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (1.18.0)

Requirement already satisfied: joblib>=1.4.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (1.5.3)

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Requirement already satisfied: pandas in /opt/conda/lib/python3.13/site-packages (from medmnist) (3.0.5)

Requirement already satisfied: scikit-image in /opt/conda/lib/python3.13/site-packages (from medmnist) (0.26.0)

Requirement already satisfied: tqdm in /opt/conda/lib/python3.13/site-packages (from medmnist) (4.67.3)

Requirement already satisfied: fire in /opt/conda/lib/python3.13/site-packages (from medmnist) (0.7.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/lib/python3.13/site-packages (from sympy>=1.13.3->torch) (1.3.0)

Requirement already satisfied: termcolor in /opt/conda/lib/python3.13/site-packages (from fire->medmnist) (3.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.13/site-packages (from jinja2->torch) (3.0.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.13/site-packages (from pandas->medmnist) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas->medmnist) (1.17.0)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (2.37.4)

Requirement already satisfied: tifffile>=2022.8.12 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (2026.7.31)

Requirement already satisfied: packaging>=21 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (26.2)

Requirement already satisfied: lazy-loader>=0.4 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (0.5)

In [2]:

```
"""
Created on Mon Aug4 17:35:36 2025

@author: vigo
"""

import os
import sys
import argparse
import time
import pickle
from tqdm import trange
import numpy as np
import PIL
import copy
import random
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
import torch.utils.data as data
import torchvision
import torchvision.transforms as transforms
from torchvision.models import vgg11, vgg16, vgg19, resnet18, resnet50
import matplotlib.pyplot as plt
from scipy.io import savemat

#Related medmnist dataset
import medmnist
```

```

from medmnist import INFO, Evaluator

# Check for Cuda
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

seed = 27
torch.manual_seed(seed)
random.seed(seed)

# Hyper-parameters
n_epochs = 10
batch_size = 128
lr = 0.0001

#Network Hyper-parameters
hidden_size = 512

data_flag = 'organamnist'
info = INFO[data_flag]
task = info['task']
n_channels = info['n_channels']
n_classes = len(info['label'])
DataClass = getattr(medmnist, info['python_class'])

#To work with 28x28 size images
#data_transform = transforms.Compose([
#    transforms.ToTensor(),
#    transforms.Normalize(mean=[.5], std=[.5])])

### To work with 224x224 resolution images
data_transform = transforms.Compose(
    [transforms.Resize((224, 224), interpolation=PIL.Image.NEAREST),
     transforms.ToTensor(),
     transforms.Normalize(mean=[0.5], std=[0.5])])

# Image Size 28x28. Set RGB =True if the dataset contains colour images
train_dataset = DataClass(split='train', transform=data_transform, download=True)
val_dataset = DataClass(split='val', transform=data_transform, download=True, as
test_dataset = DataClass(split='test', transform=data_transform, download=True,

# Loaders that give 64 example batches
train_loader = data.DataLoader(dataset=train_dataset,
                               batch_size=batch_size,
                               shuffle=True)

val_loader = data.DataLoader(dataset=val_dataset,
                             batch_size=batch_size,
                             shuffle=False)

test_loader = data.DataLoader(dataset=test_dataset,
                              batch_size=batch_size,
                              shuffle=False)

image, label = train_dataset[0] # Get one sample image
print(f"Image shape: {image.shape}")

```

Image shape: torch.Size([1, 224, 224])

```
In [3]: # select 16 random indices for plotting
indices = random.sample(range(len(train_dataset)), 16)

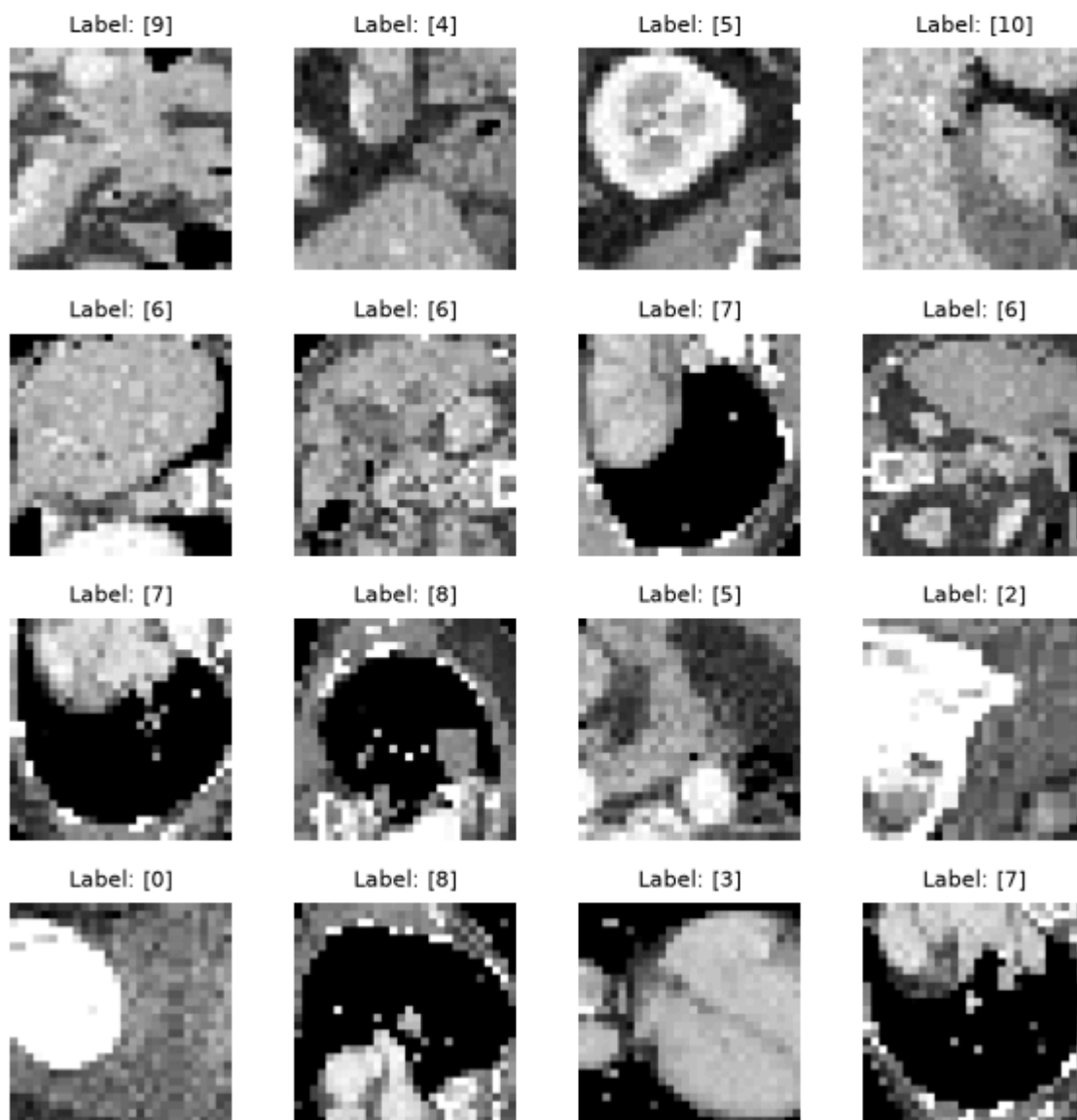
plt.figure(figsize=(6, 6))

for i, idx in enumerate(indices):
    image, label = train_dataset[idx]

    # If tensor, convert to 2D array (grayscale)
    if hasattr(image, 'squeeze'):
        image = image.squeeze() # remove channel dimension if present

    plt.subplot(4, 4, i + 1)
    plt.imshow(image, cmap='gray')
    plt.title(f"Label: {label}", fontsize=8)
    plt.axis('off')

plt.tight_layout()
plt.show()
```



```
In [7]: #VGG models
conv_model = vgg16(pretrained = True)
```

```

# adapt first conv layer to match input chnannels (this is needed if you are feed
old_conv = conv_model.features[0]
conv_model.features[0] = nn.Conv2d(
    in_channels=1,
    out_channels=old_conv.out_channels,
    kernel_size=old_conv.kernel_size,
    stride=old_conv.stride,
    padding=old_conv.padding
)

#Match the number of classes in the final fc layer
in_features = conv_model.classifier[-1].in_features
conv_model.classifier[-1] = nn.Linear(in_features, n_classes)

#Similarly Implement Transfer Learning using Resnet models.

conv_model.to(device)

# Loss and optimizer
criterion = nn.CrossEntropyLoss() #One can tray NLL loss
optimizer = torch.optim.Adam(conv_model.parameters(), lr=lr, betas=(0.9, 0.999))

```

/opt/conda/lib/python3.13/site-packages/torchvision/models/_utils.py:222: UserWarning: Arguments other than a weight enum or `None` for 'weights' are deprecated since 0.13 and may be removed in the future. The current behavior is equivalent to passing `weights=VGG16\_Weights.IMAGENET1K\_V1`. You can also use `weights=VGG16\_Weights.DEFAULT` to get the most up-to-date weights.

warnings.warn(msg)

Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to /home/uccloud/.cache/torch/hub/checkpoints/vgg16-397923af.pth

```

0%|          | 0.00/528M [00:00<?, ?B/s]
 7%|█         | 38.4M/528M [00:00<00:01, 402MB/s]
16%|██        | 82.4M/528M [00:00<00:01, 436MB/s]
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50%|██████    | 265M/528M [00:00<00:00, 472MB/s]
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69%|█████████ | 364M/528M [00:00<00:00, 493MB/s]
78%|█████████ | 411M/528M [00:00<00:00, 426MB/s]
86%|█████████ | 453M/528M [00:01<00:00, 422MB/s]
96%|█████████ | 508M/528M [00:01<00:00, 465MB/s]
100%|█████████| 528M/528M [00:01<00:00, 459MB/s]

```

```

In [8]: # Training
def train(model, dataloader, criterion, optimizer, device):
    model.train()
    train_correct = 0
    total_samples = 0

    for inputs, targets in dataloader:
        inputs, targets = inputs.to(device), torch.squeeze(targets, 1).long().to(device)

        optimizer.zero_grad()
        outputs = model(inputs)

```

```

        loss = criterion(outputs, targets)
        loss.backward()
        optimizer.step()

        _, predictions = torch.max(outputs, 1)
        train_correct += (predictions == targets).sum().item()
        total_samples += targets.size(0)

    train_acc = (train_correct / total_samples) * 100
    return train_acc, loss.item()

# Validation/Test
def val(model, dataloader, criterion, device):
    model.eval()
    val_correct = 0
    total_samples = 0
    val_loss = 0.0

    with torch.no_grad():
        for inputs, targets in dataloader:
            inputs, targets = inputs.to(device), torch.squeeze(targets, 1).long()

            outputs = model(inputs)
            loss = criterion(outputs, targets)
            _, predictions = torch.max(outputs, 1)

            val_correct += (predictions == targets).sum().item()
            total_samples += targets.size(0)
            val_loss += loss.item()

    val_acc = (val_correct / total_samples) * 100
    avg_val_loss = val_loss / len(dataloader)
    return val_acc, avg_val_loss

```

```

In [9]: Train_Acc = []
        Train_Loss = []

        Val_Acc = []
        Val_Loss = []

        for epoch in range(n_epochs):
            train_acc, train_loss = train(conv_model, train_loader, criterion, optimizer)
            val_acc, val_loss = val(conv_model, val_loader, criterion, device)
            print (f'Epoch--[{epoch+1}], Train_Accuracy: {train_acc}, Validation_Accuracy: {val_acc}')
            Train_Acc.append(train_acc)
            Train_Loss.append(train_loss)
            Val_Acc.append(val_acc)
            Val_Loss.append(val_loss)

# Plot Epoch vs Accuracy
plt.figure()
plt.plot(range(1, n_epochs+1), Train_Acc, label='Train Accuracy')
plt.plot(range(1, n_epochs+1), Val_Acc, label='Validation Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.title('Train vs Validation Accuracy')

```

```

plt.legend()
#plt.grid(True)
plt.show()

# Plot Epoch vs Loss
plt.figure()
plt.plot(range(1, n_epochs+1), Train_Loss, label='Train Loss')
plt.plot(range(1, n_epochs+1), Val_Loss, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Train vs Validation Loss')
plt.legend()
#plt.grid(True)
plt.show()

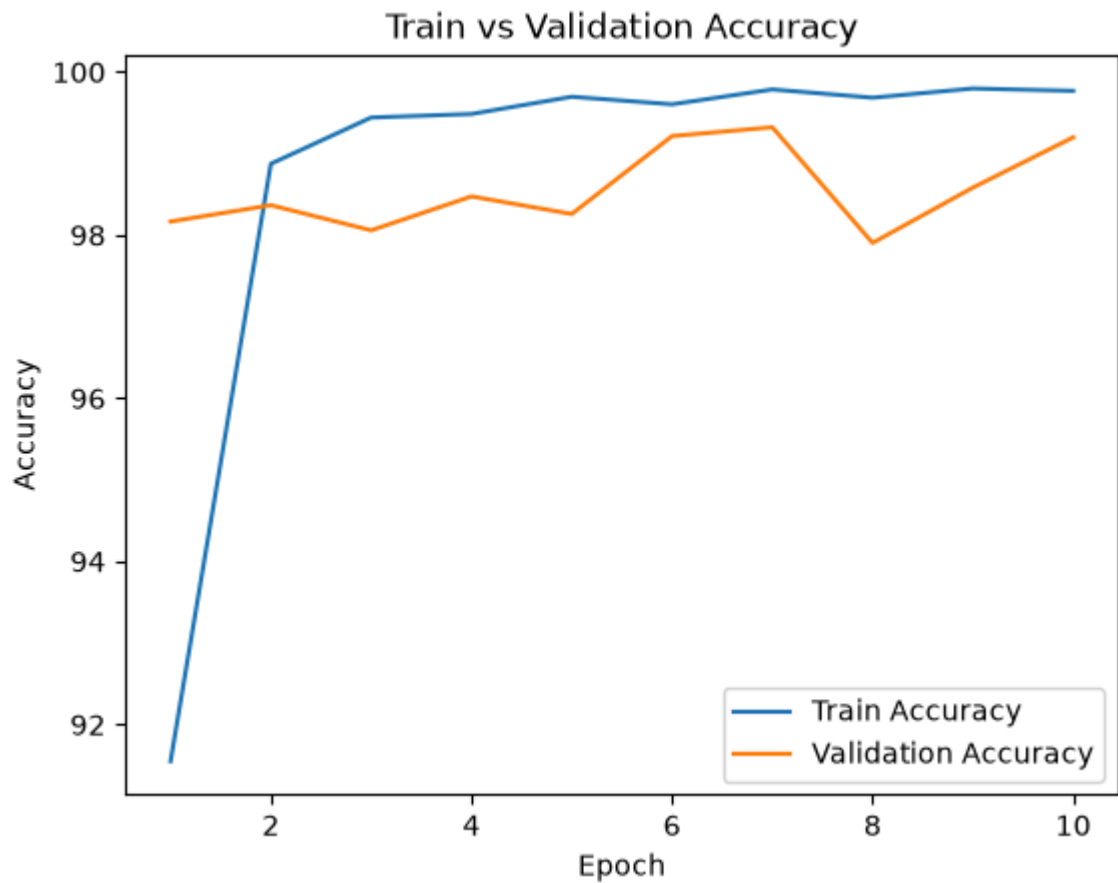
#Testing
test_acc, test_loss = val(conv_model, test_loader, criterion, device)
print (f'Test_Accuracy: {test_acc}')

```

```

Epoch--[1], Train_Accuracy: 91.55117039437516, Validation_Accuracy: 98.1666923432
4449
Epoch--[2], Train_Accuracy: 98.87445386418217, Validation_Accuracy: 98.3669696502
8501
Epoch--[3], Train_Accuracy: 99.44156708428575, Validation_Accuracy: 98.0588507163
7653
Epoch--[4], Train_Accuracy: 99.48496860623246, Validation_Accuracy: 98.4748112771
5298
Epoch--[5], Train_Accuracy: 99.69618934637307, Validation_Accuracy: 98.2591280234
1705
Epoch--[6], Train_Accuracy: 99.60359943288678, Validation_Accuracy: 99.2142967185
3336
Epoch--[7], Train_Accuracy: 99.78588582506293, Validation_Accuracy: 99.3221383454
0133
Epoch--[8], Train_Accuracy: 99.68461560718728, Validation_Accuracy: 97.9047912494
2227
Epoch--[9], Train_Accuracy: 99.79745956424873, Validation_Accuracy: 98.5826529040
2095
Epoch--[10], Train_Accuracy: 99.76852521628426, Validation_Accuracy: 99.198890771
83794

```



Test_Accuracy: 91.98447519406008

In []: